

MATH 220/199 Exam 2 Mini Review

Reminders

1. Office hours are held in Davenport 330 at 3pm on Thursdays.
2. Do the Merit Quiz on the Merit Canvas page.

Definitions

Derivative Rules:

1. $[f(x)g(x)]' =$
2. $[f(g(x))]' =$
3. $\left[\frac{f(x)}{g(x)}\right]' =$
4. $[f(x)^{g(x)}]' =$ (This one will take some work to find a general form, and you may want to first define $h(x) = f(x)^{g(x)}$)

Laws: For each law, state both the conditions necessary to apply the law, and the conclusion the law allows you to make:

1. Mean Value Theorem:
2. L'Hôpital's Rule:

Other Definitions: State definitions for the following terms:

1. Critical Point:
2. Inflection Point:
3. Local Minimum/Local Maximum:
4. Absolute Min/Max

General Practice

1. Give the set of critical points for the implicit function $\sin(xy) + \arctan(xy) = 1$
2. Compute the inflection points (x, y) for $h(x) = x^4 - 6x^3 + 8x^2 + 6x - 9$
3. CHALLENGE: Let a, b, c be real numbers and let $f(x) = ax^2 + bx + c$. Find all such (a, b, c) so that:
 - (a) $f(x) > 0$ for all x
 - (b) $f(x)$ is everywhere concave up
 - (c) $f(x)$ has a critical point at $x = 2a$